

Chapter 117 — Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

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PREPARING THE PATIENT FOR PROLAPSE SURGERY

- Pelvic organ prolapse (POP) is **largely a quality of life (QoL) condition**, so the decision to operate rests on the interaction of prolapse grade, symptoms and the degree of QoL impairment — not on anatomy alone.
- Procedure selection is **individualised to the clinical presentation**: physiological age, comorbidities, previous surgery, level of physical and sexual activity, and overall bother.
- **The first operation offers the best chance of restoring normal support and function.** After it the normal anatomical planes no longer exist, adding difficulty to any subsequent repair.
- **Recurrence rates rise with each successive attempt** at surgical correction.
- The surgeon must understand **the anatomical alterations each technique produces**, and how the operation will affect the continuity and function of the endopelvic fascia and its related viscera.
- Preoperative discussion of risks and benefits allows the patient to choose the operation — **or to decline surgery altogether, which is a legitimate informed decision** where bother does not justify the risk.

Classification of Prolapse Repairs

Three categories, defined by what they do to the native tissue:

- **Obliterative** — for patients whose comorbidities preclude prolonged surgery; **symptom relief with minimal morbidity. Contraindicated if the patient wishes to remain sexually active.**
- **Restorative** — for **discrete defects in the fibromuscular layer of the endopelvic fascia**; corrects the defect using endogenous support structures.
- **Compensatory** — for **weak native tissue or previous failed surgery**; reinforces the repair with a synthetic, biological or autologous graft.

Route of Reconstruction

- Reconstruction may be **vaginal, abdominal (open, laparoscopic or robotic), or a combination.**
- Because **POP frequently affects multiple compartments**, surgery often combines repairs, and **an anti-incontinence procedure may be performed concurrently.**
- **Vaginal**: fewer complications, shorter recovery, shorter operative time, shorter hospital stay.
- **Abdominal**: greater durability of repair.
- **Open versus laparoscopic**: complication rates equivalent, but recovery better with the minimally invasive approaches, making them progressively more attractive.

Preoperative Counselling Before Vaginal Prolapse Surgery

- The definition of surgical success is evolving, because **success and objective resolution of the anatomical defect are not the same thing.**
- **Absence of symptoms — resolution of vaginal bulge and minimisation of de novo postoperative symptoms — correlates most strongly with the patient's own assessment of cure.** Success must therefore incorporate patient satisfaction and symptom improvement alongside objective measures.
- **The absence of symptoms influences satisfaction more than the elimination of anatomical prolapse.**

- **The process begins preoperatively:** expectations and readiness for surgery demonstrably affect later satisfaction and perception of improvement.
- **Satisfaction correlates highly with achievement of the patient's own self-described goals** — hence the value of asking directly what she expects and what changes she wants.
- **Patients misinterpret routine postoperative events as complications** — pain, discharge home with a catheter, constipation, urgency incontinence, minor anaesthetic effects. These perceived complications drive dissatisfaction **even where objective cure rates are high.**
- **Feeling unprepared for surgery was strongly associated with a negative perception of outcome.**
- After a **three-page handout on what to expect**, women who felt fully prepared were **more likely to be improved on the Patient Global Impression of Improvement (PGI-I)** and had **lower Pelvic Organ Prolapse Distress Inventory (POPDI)** and **Urinary Distress Inventory (UDI)** scores. **Objective measures of cure did not differ by preparedness** — preparation changes perception, not anatomy.

The consent discussion, in sequence

- Alternatives to surgery → the purpose of the planned operation, what it can and cannot achieve → the benefits, which symptoms can improve → risks and complications → what to expect in hospital → what to expect at home → strategies for coping with a catheter in both settings.

KEY POINTS: PREOPERATIVE COUNSELLING FOR PELVIC ORGAN PROLAPSE SURGERY

- **Absence of recurrent POP symptoms influences patient satisfaction more than elimination of anatomical prolapse.**
- **Understand the patient's goals and expectations**, and review the risks and benefits of each surgical approach carefully.
- **Postoperative "events" — catheterisation, urinary tract infection (UTI) — are significant to the patient** and negatively influence perception of outcome.
- **The informed consent discussion must cover goals and expectations alongside risks and benefits** of each surgical option.
- **Educate the patient on the whole perioperative course**, because perioperative events are otherwise perceived as complications and degrade overall perception of the procedure.

SURGICAL MANAGEMENT OF PELVIC ORGAN PROLAPSE

TABLE 117.1 COMPARTMENTAL ANATOMY: PELVIC ORGAN PROLAPSE QUANTIFICATION (POP-Q) SITES AND ORGAN INVOLVEMENT

Compartment	POP-Q site	Prolapsed organ	Vaginal wall site
Anterior	Aa	Urethrocele	Distal anterior vaginal wall
Anterior	Ba	Cystocele	Proximal anterior vaginal wall
Middle	C	Cervix / vaginal cuff	Cervix / vaginal cuff
Middle	D	Enterocoele	Uterosacral scar
Posterior	Ap	Rectocele	Distal posterior vaginal wall
Posterior	Bp	Enterocoele	Proximal posterior vaginal wall

Compartment	POP-Q site	Prolapsed organ	Vaginal wall site
Posterior	—	Perineal body defects	Perineal body

- **The A points are always distal, the B points always proximal**, on both walls — the pairing that makes the table memorable.
- **Point D exists only where the cervix is present**; after hysterectomy it is omitted and the uterosacral scar is the landmark.
- **The trap** is the posterior pair, which inverts intuition: **Ap is distal (rectocele), Bp proximal (enterocele)**. Anchor them to the anterior pair rather than to the organ.

TABLE 117.2 SURGICAL APPROACH TO PELVIC ORGAN PROLAPSE

POP-Q site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy
Ba — bladder	Anterior repair; paravaginal repair; colpocleisis; wedge colectomy	Paravaginal repair; abdominal sacrocolpopexy
C — cervix / cuff	Uterosacral ligament suspension; iliococcygeus suspension; sacrospinous fixation; Manchester operation; hysteropexy; vaginal hysterectomy; colpocleisis	Abdominal hysterectomy; uterosacral ligament suspension; abdominal sacrocolpopexy; uterine suspension
D — cul-de-sac	McCall culdoplasty	Halban culdoplasty; Moschcowitz culdoplasty
Ap — posterior	Rectovaginal plication (posterior repair); site-specific repairs	Colpoperineopexy

- **Paravaginal repair and uterosacral ligament suspension appear in both columns** — the only two operations performable by either route.
- **McCall is the vaginal cul-de-sac operation; Halban and Moschcowitz are the abdominal ones.** The table sets exactly that swap as its trap.

Anterior Compartment

Anterior Colporrhaphy

- **Kelly (1913) first described cystocele repair as a treatment for urinary incontinence**, emphasising plication of the pubocervical fascia to correct the central defect.
- The operation became the **anterior colporrhaphy, or native-tissue cystocele repair**, and is now used for **anterior prolapse rather than for incontinence**.
- **Its use for incontinence collapsed** after the American Urological Association (AUA) Female Stress Urinary Incontinence (SUI) Clinical Guidelines Panel meta-analysis showed **failure approaching 40%**.

Central versus lateral defect

- **Central defect** = compromise of the **pubocervical fascia in the midline**, producing an anterior bulge.
- **Lateral (paravaginal) defect** = **separation of the pubocervical fascia from its attachment to the arcus tendineus fasciae pelvis (ATFP)**.
- **Anterior compartment defects are commonly combined central and lateral**, so **anterior colporrhaphy usually must be combined with a paravaginal repair or lateral anchoring**.
- **Isolated central defects are rare**; only there does colporrhaphy alone suffice.

Why the reported failure rates conflict

- Early series reported high objective success; a later randomised trial of three techniques reported a high failure rate for native tissue repair, and re-analysis of that same dataset with a contemporary definition gave a substantially better anatomical result.
- **The spread in the literature is explained by two things: variability in the definition of failure, and the fact that early series operated on patients with multiple uncorrected defects.**
- **Recurrence rates today may be no better than those of 150 years ago**, owing to variation in evaluation, technique and follow-up — **even between urogynaecologists within one practice.**

Anterior Colporrhaphy: Technique

- **Dorsal lithotomy**, all pressure points padded, **hip and knee flexed to approximately 90 degrees** → **preparation and cleansing** of vaginal and perivaginal tissue → **retraction, fixed or hand-held**, by body habitus and available assistance →
- **Indwelling urethral catheter; bladder drained continuously or intermittently** → **hydrodissection**, with or without a vasoconstricting agent → **midline incision in the anterior vaginal wall, from the vaginal apex to the bladder neck** →
- **Dissect the vaginal wall off the pubocervical fascia, midline first and then laterally to the ATRP** — the dissection must expose enough to **delineate both central and lateral defects** → **exposure with Allis clamps or a self-retaining ring retractor** →
- **Reduce the bladder with finger or instrument**, permitting lateral exposure and reapproximation of the lateral pubocervical tissues in the midline → **interrupted 2-0 or 3-0 delayed absorbable plication sutures placed sequentially from the bladder neck to the vaginal apex** → **tie while the assistant reduces the prolapsed tissue** →
- **Trim excess anterior vaginal wall judiciously** → **close with running absorbable suture** → **intravenous dye and cystoscopy** to exclude iatrogenic bladder injury and confirm ureteric patency and efflux → **vaginal packing** to cover the field and assist haemostasis.

Incision planning where a sling is performed concurrently

- **Midurethral sling: do NOT extend the colporrhaphy incision to the urethra.** Place the sling through a separate midurethral incision.
- **Pubovaginal sling: DO create a single incision from the vaginal apex to the midurethra**, allowing precise positioning over the bladder neck.

Safety point

- **Avoid excessively deep suture placement**, which may **penetrate the bladder or ureter, or kink the distal and intramural ureter** — the rationale for mandatory completion cystoscopy.

Figure 117.2 — Anterior colporrhaphy. The **anterior fibromuscularis layer is imbricated with 2-0 delayed absorbable suture, continuous or interrupted.** (From Nichols DH: *Cystocele*. In Nichols DH, editor: *Gynecologic and Obstetric Surgery*. St Louis: Mosby; 1993, pp 334–362.)

Results of Anterior Colporrhaphy

- **Earlier studies reported cure rates approaching 100%**; large retrospective and randomised series from that era gave very low recurrence.
- A patient-satisfaction study found **objective recurrence considerably exceeded subjective failure**, demonstrating that **objective anatomical recurrence does not correlate negatively with overall patient satisfaction.**
- **Anterior prolapse recurrence is roughly halved by a concomitant sling.**

- **In series comparing colporrhaphy against augmented repairs, recurrence approaches 40% —** markedly higher than the older series, **largely because of alternative and composite outcome measures**, not a change in the operation.

Augmented anterior repair — the randomised evidence

- **Sand trial** — anterior colporrhaphy with **polyglactin 910 (Vicryl) suture alone** versus **colporrhaphy plus a free Vicryl mesh inlay placed beneath the trigone**. At one year, **failure 25% with the inlay versus 43% with suture plication alone (P = 0.02)**.
- **Weber trial** — three arms in symptomatic cystocele (predominantly **POP-Q stages 2 and 3**): colporrhaphy reinforced with Vicryl mesh, traditional plication without tension using polydioxanone suture, and **"ultralateral" plication with polydioxanone and tension**. With failure defined as **POP-Q stage 2 or worse**, failure was **30%, 46% and 42% — no significant difference**, and the trial was **underpowered**. Its inference is that **no suturing technique is superior to another**.
- The same investigators had earlier reported recurrence after standard colporrhaphy of only **20%** — a discrepancy generated purely by the outcome definition.
- **The Sand and Weber trials are not sufficiently similar to permit meta-analysis.**
- **Chmielewski re-analysis** of the Weber dataset, failure redefined as **recurrent prolapse beyond the hymen: 11%, 23% and 9%**; **combined objective recurrence 12%**, **no significant difference between groups**.
- **It was the pre-re-analysis figures** — apparent high recurrence and poor durability in moderate- and high-grade prolapse — **that drove investigators towards augmented repair**.

Graft and mesh evidence

- Long-term randomised follow-up of colporrhaphy versus ultralateral colporrhaphy with dermal graft found **no difference in success at 7 to 10 years**.
- **Cochrane review (2016), 33 trials: biological graft repair or absorbable mesh confers minimal advantage over native tissue repair**. However, **against polypropylene mesh, native tissue repair was associated with greater awareness of prolapse, higher rates of repeat prolapse surgery, and more anterior compartment recurrence**.
- A large retrospective cohort found **biological grafts had higher recurrent POP, but graft complications were far higher with synthetic grafts**.
- **This asymmetry — synthetics work better but complicate more — together with the FDA 522 post-market surveillance studies, justifies the withdrawal of transvaginal synthetic mesh for POP repair from the global market.**

TABLE 117.3 RESULTS OF ANTERIOR COMPARTMENT REPAIR WITHOUT GRAFT

Comparison	Follow-up	Result
Anterior colporrhaphy (AC)		
AC, pre-2000 series	1.2–20 yr	Success 80–100%
AC vs Burch colposuspension	8–17 yr	97 vs 66%
AC vs AC + Vicryl inlay (Sand)	1 yr	57 vs 75%; no mesh complications
AC vs AC + Vicryl mesh (Weber)	23 mo	37 vs 42%; no mesh complications
AC vs AC + dermal graft	7–10 yr	56 vs 68%
	74 mo	

Comparison	Follow-up	Result
AC vs AC + posterior colporrhaphy		68% subjective; no prolapse symptoms among objective recurrences
Vaginal paravaginal repair (VPR)		
VPR series	0.5–3 yr	Success 67–100%
AC + VPR vs AC alone	13 vs 24 mo	54 vs 45%
Abdominal paravaginal repair (APR)		
APR series	0.5–6 yr	Success 95–97%
APR + sling vs APR alone	17 mo	93 vs 76%
APR vs VPR	10 yr	89 vs 70%
APR vs VPR vs laparoscopic	12 mo	APR and VPR similar; laparoscopic arm abandoned for low success
APR vs AC + mesh	2 yr	65 vs 67%
Anterior colporrhaphy with sling		
AC + sling vs AC alone	1 yr	81 vs 58%

- **Success falls as follow-up lengthens and outcome definitions tighten** — pre-2000 series cluster at 80–100%, post-2000 series at 37–68% for the same operation.
- **Graft or inlay augmentation gives a modest and inconsistent gain**, favourable at one year and marginal by two.
- **Adding a sling produces the largest single improvement** in anterior compartment success — larger than any graft effect in the table.
- **Abdominal paravaginal repair outperforms the vaginal route at long follow-up while the two are equivalent at one year** — durability, not early success, is the abdominal advantage.

Complications of Anterior Colporrhaphy

- **Reported complications:** de novo or worsened SUI; de novo overactive bladder symptoms; postoperative urinary retention; incomplete bladder emptying; **significant bleeding, greater than 350 mL**; wound infection; bladder or ureteric injury; vesicovaginal fistula; vaginal shortening; epithelial inclusion cyst; dyspareunia.

Detrusor overactivity

- **De novo detrusor overactivity occurs in 5–7%** after standard colporrhaphy.
- **Pre-existing detrusor overactivity resolves in up to 63%** after surgical repair of POP.

De novo stress urinary incontinence — CARE trial secondary analysis

- The incidence of de novo SUI correlates directly with the degree of cystocele on preoperative examination.
- By POP-Q stage of anterior prolapse: stage 2 → 41.3%; early stage 3 → 52.5%; advanced stage 3 or stage 4 → 66.1%.
- **Point Ba was significantly associated with de novo SUI** on univariate (odds ratio 1.17, P = 0.015) and multivariate analysis (odds ratio 1.16, P = 0.04).

Urinary retention — mechanism and management

- Usually follows a concomitant anti-incontinence procedure, but may occur without one, from neurological trauma and impaired bladder contractility caused by the plication itself.
- Most retention resolves after a brief period of clean intermittent catheterisation (CIC) → persistent retention after a concomitant anti-incontinence procedure → sling incision or urethrolisis → impaired emptying from pre-existing hypocontractility → chronic CIC, then neuromodulation, then suprapubic tube drainage as a last resort.

Bleeding

- Two sources: the high vascularity of the vagina, and the venous complex beneath the inferior pubic bone.
- Problematic bleeding indicates dissection in the wrong plane. The vaginal wall must be dissected off the pubocervical fascia directly on its white, shiny surface.
- Repeat anterior colporrhaphy carries higher bleeding risk because scarring obscures the plane.

Bladder and ureteric injury

- Rare in standard anterior colporrhaphy. Bladder injury occurs on perforating into the retropubic space or during dissection of the vaginal flaps, especially with atrophic tissue. Ensure proper bladder drainage before perforating the endopelvic fascia.
- Inadvertent bladder injury → two-layer closure with absorbable suture. Prior pelvic irradiation or a tenuous repair → consider labial fat pad interposition (Kreder) to prevent vesicovaginal fistula.

Confirming ureteric efflux

- Indigo carmine — blue (intravenous); fluorescein 10% — green (intravenous); phenazopyridine — orange (preoperative, oral). National shortages of indigo carmine drove adoption of the alternatives.
- If stained urine does not efflux from both orifices → ureteric catheterisation, retrograde pyelography, or removal of the plication sutures.
- A patent ureter may still fail to efflux — consider inadequate fluid resuscitation or ureteric kinking. If the ureter cannot be catheterised, take whatever steps are needed to ensure patency.

Concept — *The anterior compartment's entire literature is a dispute about definitions, not about operations. One randomised dataset yields 30–46% failure or 9–23% depending on whether the threshold is POP-Q stage 2 or descent beyond the hymen — so every quoted recurrence rate is uninterpretable without its definition, and the apparent superiority of augmentation shrinks as the definition is made clinically meaningful.*

PROCEDURES FOR LATERAL AND COMBINED DEFECTS**Vaginal Paravaginal Repair**

- George White first described the vaginal paravaginal repair: a cystocele repair suturing the lateral vaginal sulci to the "white line", the ATRP.
- White's premise: a cystocele results from weakness of the lateral vaginal attachments to the ATRP — the lateral, not the central, hypothesis.
- White's concept fell into disfavour for roughly 70 years while Kelly's anterior colporrhaphy dominated.
- Richardson (1976) reintroduced the paravaginal defect repair, but by an abdominal approach.
- Whatever the route, the goal is to repair the lateral defect by reattaching the pubocervical fascia to the ATRP, and the repair may be combined with anterior colporrhaphy where lateral and central defects coexist.

Vaginal Paravaginal Repair: Technique

- **Midline vertical incision through the anterior vaginal wall, bladder neck to vaginal apex → sharply dissect the vaginal wall off the attenuated pubocervical fascia and bladder → carry the dissection laterally to the pelvic sidewall, blunt dissection permissible → perforate the endopelvic fascia to access the sidewall →**
- **On entering the retropubic space, palpate to identify the ATRP as it courses from the ischial spine to the inferior aspect of the pubic ramus → deep lighted retractors → retract the bladder medially and drain it adequately →**
- **Place five to seven interrupted non-absorbable sutures at 1 cm intervals through the ATRP, using the Capio suture-capturing device in the deep retropubic space → pass them through the lateral edge of the detached pubocervical (vesicopelvic) fascia → tie no suture until all are placed, on one side or both →**
- **If indicated, place central defect plication sutures now to complete a combined repair → cystoscopy to confirm ureteric patency and absence of intravesical sutures → trim the epithelium judiciously and close with 2-0 absorbable suture → vaginal packing.**

Results of Vaginal Paravaginal Repair

- **Most patients are free of any evidence of prolapse at short follow-up, and of the recurrences, only a minority reach or pass the hymen, all of lesser degree than preoperatively.**
- **In grade 3 anterior wall prolapse, lateral recurrence was 8% and central recurrence 22%. Midline central defects proved common, which is what drove surgeons to combine the repair with anterior colporrhaphy.**
- **In grade 2–4 combined cystocele treated by paravaginal repair plus concomitant anterior colporrhaphy, lateral recurrence was 2% and central recurrence again 22%.**
- **In grade 2–4 prolapse, central recurrence was 3%, but of those with concomitant SUI, 57% had persistent SUI after the repair.**
- **The paravaginal repair, like standard anterior colporrhaphy, is ineffective as a treatment for stress urinary incontinence.**
- **Anterior colporrhaphy alone versus colporrhaphy plus paravaginal repair showed no significant difference in subjective or objective outcome.**

Complications of Vaginal Paravaginal Repair

- **Infrequent, but frequently significant when they occur: major intraoperative haemorrhage; lower limb neuropathy after prolonged lithotomy positioning; transfusion rate 12%; vaginal abscess requiring drainage; retropubic haematoma requiring re-exploration; bilateral ureteric obstruction that resolved only after takedown of the repair and replacement of the sutures.**
- **Technically more demanding than standard anterior colporrhaphy, with higher and more serious complication rates.**
- **The conventional repair relies on suture placement through weakened pubocervical fascia — it anchors to the tissue that failed. To compensate, most surgeons now correct lateral defects transvaginally using grafts attached bilaterally to the pelvic sidewalls.**

Concept — The paravaginal repair reliably controls the lateral defect (2–8% recurrence); the central defect recurs at around 22% regardless of whether colporrhaphy is added. The compartment is therefore treated as two independent failures, not one.

OTHER PROCEDURES TO CORRECT ANTERIOR COMPARTMENT DEFECTS

Abdominal Paravaginal Defect Repair: Technique

- Position and prepare for both abdominal and vaginal access; drain the bladder → enter the retropubic space through a low midline or transverse incision, laparoscopic or robotic equally acceptable → identify bladder neck, symphysis pubis, endopelvic fascia, ATRF and obturator fascia →
- Identify the site of normal vaginal attachment on the pelvic sidewall, from the internal aspect of the superior pubic ramus to the ischial spine → place the non-dominant hand in the vagina and elevate the lateral superior vaginal sulcus to its normal attachment along the ATRF →
- With the bladder retracted medially, place four to six interrupted non-absorbable sutures at 1 cm intervals through the ATRF, from ischial spine to pubic bone → pass these into the corresponding position in the lateral vaginal wall → avoid the paravaginal veins →
- Tie, then perform cystoscopy for ureteric patency and absence of intravesical sutures → close; leave the catheter indwelling.

Suture counts

- Vaginal paravaginal repair: five to seven sutures. Abdominal paravaginal repair: four to six sutures. Both non-absorbable, at 1 cm intervals along the ATRF between ischial spine and pubic bone; **both mandate completion cystoscopy.**

Anterior Compartment Repair With Sling

The Mechanism

- In high-stage anterior compartment prolapse, descent creates urethral kinking and compression. This compression prevents leakage despite poor urethral function, so stress incontinence is unmasked once the prolapse is corrected.
- SUI occurring in a presumably continent woman after prolapse correction is termed stress incontinence on prolapse reduction — formerly "occult SUI".
- Every patient with high-stage prolapse must be evaluated with the prolapse reduced.
- Across four prospective studies, prolapse repair in clinically continent women produced postoperative SUI in 8.3–66.1% — the width of that range is itself the point.

Methods of Reducing the Prolapse for Testing

- Options: vaginal packing, rectal swabs, speculum blade, or pessary.
- Vaginal pack of two rolled 4 × 4-inch gauzes → occult SUI in 69%, with superior visualisation of the vesicourethral angle during fluoroscopic urodynamics.
- Gauze packing plus speculum blade in high-stage prolapse → occult SUI in at least 50%.
- Fitted vaginal pessary → occult SUI in 58%, with no urethral obstruction caused by the pessary.
- Comparing rectal swabs, Gellhorn pessary and Graves speculum blade, rectal swabs were judged superior because reduction with swabs revealed a significantly lower midurethral closure pressure.
- Swab reduction gave the highest positive predictive value (79%); the speculum identified the most SUI (30%) versus only 6% with pessary; overall reduction tests had low sensitivity and high specificity.
- The techniques are not standardised, which is why the reported incidence of occult SUI varies so widely. No method has been proven superior, and preoperative reduction during urodynamics cannot reliably predict postoperative SUI.

Concomitant Versus Staged Sling

COMPARISON: THE TWO POSITIONS ON A PROPHYLACTIC SLING

Question	For a concomitant sling	For a second-stage sling
Core argument	The diagnosis is non-reproducible , so occult SUI may simply be missed	Sling complications equal the risk of the SUI they prevent
Supporting evidence	No correlation between preoperative urethral position under any reduction method and its position after surgery	Intervention for sling obstruction 8.5% vs intervention for postoperative SUI without a sling 8.3%
Practical conclusion	Perform an anti-incontinence procedure at high-stage prolapse repair	Prophylactic sling does not reduce the need for a second operation where SUI cannot be demonstrated

- The protocol that generated the staged position: **urodynamics in all patients; if SUI → concomitant midurethral sling. If no SUI → urodynamics repeated with the prolapse reduced by pessary; if occult SUI → sling.**
- In those with clinical SUI symptoms but no urodynamic and no reduction-demonstrated SUI, the risk of intervention was **30%** — an intermediate group at genuine risk.

Prediction Models for Postoperative Stress Urinary Incontinence

COMPARISON: PREDICTION MODELS FOR DE NOVO STRESS INCONTINENCE

Model	Predictors	Performance	Limitation
OPUS	Seven	Prediction accuracy 0.72 vs 0.54 for preoperative stress test alone; better than subspecialty experts	Poorer for de novo SUI; applies only to women without preoperative SUI symptoms
CUPIDO	Nine	Urodynamics adds no predictive value in asymptomatic patients ; overall 28% risk of postoperative SUI	Not generalisable to abdominal prolapse surgery
Oh	Six	Area under the curve 0.74 (0.62–0.86) vs CUPIDO 0.63 (0.49–0.76)	Validated only against CUPIDO

- OPUS = Outcomes following vaginal Prolapse repair and mid-Urethral Sling. CUPIDO = Concomitant surgery and Urodynamics investigation in genital Prolapse and stress Incontinence: A Diagnostic study including Outcome evaluation.
- Oh's six predictors: **age, diabetes mellitus, subjective urinary incontinence, prolapse reduction stress test result, type of prolapse surgery, concomitant midurethral sling.**
- Practical output: **patients with subjective SUI undergoing native tissue repair or colpocleisis do not carry a greater than 50% risk of bothersome postoperative SUI** — so a concomitant anti-incontinence procedure may reasonably be omitted.

Sling as an Adjunct to Anterior Compartment Support

- A concomitant suburethral sling reduced postoperative cystocele recurrence by **55%** — the sling contributes to the durability of the anterior repair itself, not only to continence.
- Simultaneous pubovaginal sling improved support in grade 3 and 4 cystoceles.

The Randomised Trials

COMPARISON: RANDOMISED TRIALS OF CONCOMITANT ANTI-INCONTINENCE SURGERY

Trial	Population and comparison	Result	Verdict
CARE	Stage 2–4 POP, abdominal sacrocolpopexy ± Burch colposuspension	SUI 44.1% → 23.6% at 3 months; 32% vs 45% at 2 years	First large randomised trial of the question; one third still incontinent after Burch
E-CARE	Long-term extension of CARE	Probability of SUI at 7 years 62% vs 77% ; time to treatment failure longer with urethropexy	Benefit is delay and attenuation, not prevention
OPUS	Stage 2 or worse POP without SUI symptoms, vaginal surgery ± midurethral sling vs sham	Incontinence 27% vs 43% at 12 months ($P = 0.002$); more bladder perforation, UTI, bleeding and incomplete emptying with the sling ($P \leq 0.05$)	Sling works but costs harm; no difference in patient-reported pelvic floor symptoms

Screening and Selective Use

- There is no standardised test for occult SUI.
- All patients with advanced-stage anterior compartment prolapse should be screened with the prolapse reduced, and a concomitant anti-incontinence procedure considered.
- Those with preoperative SUI symptoms, occult SUI or urodynamically proven SUI are at elevated risk and should consider a simultaneous anti-incontinence procedure.
- In continent women the risks and benefits must be reviewed individually. The literature supports selective, not routine, use.

KEY POINTS: ANTERIOR COMPARTMENT REPAIR

- Anterior compartment defects are central, lateral, or combined. Central = midline separation or attenuation of the pubocervical (perivesical) fascia. Lateral = detachment of the pubocervical fascia from the ATFP.
- Most anterior compartment defects have concomitant lateral and/or apical defects. Isolated anterior colporrhaphy is not recommended when additional defects are present.
- Anterior compartment repair does not suspend the vaginal apex. Coexisting anterior and apical prolapse requires concomitant repairs addressing each defect separately.
- Grafts improve anatomical results, but graft complications occur. Subjective results are unchanged by biological grafts and improved by synthetic grafts. QoL measures, reoperation for prolapse and dyspareunia rates are unchanged by synthetic mesh. Despite the anatomical evidence, most mesh products and many biologics are no longer available.
- Anterior colporrhaphy and paravaginal repair are both ineffective alone in the treatment of SUI.
- All women with advanced-stage anterior compartment prolapse should be screened for occult SUI with prolapse reduction, and the risks and benefits of a concomitant anti-incontinence procedure discussed.

APICAL COMPARTMENT

- Middle compartment defects commonly involve the vaginal apex, with uterus, bowel or omentum, and — depending on defect size — bladder and rectum.

- **The apex is the cornerstone of vaginal support. Failure to secure apical support at the time of prolapse correction increases recurrence risk exponentially.**
- **Correction may be vaginal, abdominal or minimally invasive, with or without uterine preservation.**

Apical Vault Prolapse Repair via Uterosacral Ligament Suspension

- **Uterosacral ligament suspension preserves the natural orientation of the vaginal axis, potentially preventing recurrence in other segments.**
- **The uterosacral ligaments break at specific points rather than attenuating diffusely across the uterosacral-cardinal complex — this is what permits a site-specific repair.**
- **Counterpoint:** histological assessment questions the integrity of the structure for long-term apical support.

Surgical Anatomy of the Uterosacral Ligaments

- **Attachment:** broadly to S1–S3, variably to S4. **Course:** fanlike, anterolaterally, to the cervical os and onto the proximal posterior vagina. **Three portions:** sacral, intermediate, cervical.
- **The superior gluteal vein runs beneath the sacral portion.**
- **Position relative to the ischial spine is highly variable; the ureter lies near the anterior margin of the ligament close to the cervix, and the ligament-to-ureter distance narrows from 4 cm near the sacrum to 0.9 cm near the cervix.**
- **The fibrous ureteric sheath makes the ureter liable to kinking from traction by sutures in adjacent tissue — kinking, not ligation, is the mechanism.**
- **Sutures close to the sacrum risk entrapping fibres of the sacral plexus trunk (S1–S4), producing buttock pain radiating to the posterior thigh and popliteal fossa.**
- **Cadaveric work:** the ischial spine is the most useful pelvic landmark, and S3 nerve injury increases with suture placement beyond a mean of 4.6 cm from it, cephalad along the ligament.

The safe zone — three rules

- **The intermediate portion is the optimal suture site:** fewest structures at risk, stable fixation, and furthest from the ureter and its sheath, so least traction transmitted to it.
- **Locate the intermediate segment 1 cm posterior along the palpable ligament at the level of the ischial spine, with the ligament on tension.**
- **Place sutures lateral to medial to avoid catching the ureter or its attachments.**
- **Advantage over sacrospinous fixation:** less injury to the pudendal and gluteal vessels.

High Uterosacral Vaginal Vault Suspension: Technique

- **The procedure begins intraperitoneally → after hysterectomy, identify the apex on the vaginal cuff at the 3 and 9 o'clock positions, the attachments of the cardinal and uterosacral ligaments → place stay sutures at these dimples before incision →**
- **Dissect the vaginal epithelium away from the peritoneum of the hernia sac; mobilise the sac to its neck, enter it, excise it → pack the bowel away with laparotomy pads → palpate the ischial spines transperitoneally; the ligaments lie posterior and medial to them →**
- **Apply traction to the stay sutures, or place an Allis clamp at the 5 and 7 o'clock positions, to tent up the ligament → place 2 to 4 non-absorbable sutures through the uterosacral-cardinal complex → confirm ureteric position — the ureters should lie lateral, but their position relative to the ligament edge varies significantly with the degree of prolapse; place temporary stents to aid palpation if in doubt →**

- **Pass needles lateral to medial, exiting away from the ureter → pass the suture at the level of the ischial spines for correct depth; place each subsequent suture proximal to the last → perform anterior colporrhaphy or sling now, if planned →**
- **Place the suspensory sutures into the most apical portions of the pubocervical and rectovaginal fascia — highest sutures most medially, more distal sutures most laterally — and pass them out through the vaginal epithelium on each side →**
- **Cystoscopy with indigo carmine, performed BEFORE trimming the sutures — untrimmed tails allow the offending suture to be found — and AFTER the suture is tied, not merely placed, because the obstruction is created by tying → if no efflux, remove sutures beginning with the most lateral on the ipsilateral side, then work through the remainder in sequence.**
- **As the suspensory sutures are tied, the rectovaginal and pubocervical fascia are drawn together at the ligament complex — this resuspends the apex and closes the cul-de-sac simultaneously.**

Extraperitoneal Approach

- **Midline incision, its extent determined by the concomitant prolapse → dissect vaginal epithelium off the endopelvic fascia; identify the enterocele sac and free it from the vault → continue laterally until the cardinal-uterosacral complex is identified posterior and medial to each ischial spine → place tension on the vagina to tent the complex for palpation →**
- **Use a Breisky-Navratil retractor to retract bladder and ureters anteriorly — this is what prevents ureteric obstruction from traction, or injury → place sutures as for the intraperitoneal approach; cystoscopy and vaginal closure as described.**

Abdominal Approach to the Uterosacral Ligaments

- **Begin by elevating the uterus at the time of abdominal hysterectomy → place three permanent sutures in each ligament, proximal and medial to the ischial spine, numbered in Shull's manner → complete the hysterectomy and close the cuff →**
- **Pass the sutures sequentially through the anterior and posterior leaves of the endopelvic fascia → reapproximating the anterior and posterior vaginal muscularis closes any potential enterocele defect and elevates the cuff towards the sacrum, recreating the normal vaginal axis →**
- **Cystoscopy after tying; leave sutures uncut until efflux is demonstrated → if no efflux, remove the most lateral ipsilateral suture, then take down subsequent sutures in sequence.**
- **Suture counts: vaginal 2 to 4 non-absorbable sutures per side; abdominal 3 permanent sutures per side.**

Results of Uterosacral Ligament Suspension

- **The transvaginal evidence base is predominantly uncontrolled retrospective case series: mean objective success 85%, range 48–96%; mean reoperation rate 5.8%.**
- **Meta-analysis by compartment: apical 98.3%, posterior 87.4%, anterior 81.2%. Where postoperative symptoms were reported, 82–100% obtained relief. Ureteric kinking or injury 1–11%.**
- **The largest series: 87% had no recurrent support defect at any site on any postoperative examination; only 5% had grade 2 or worse prolapse, and the commonest site of recurrence was the anterior compartment. One patient required ureteroneocystostomy; two required removal of suspensory sutures.**
- **Another large series: 89% satisfaction; 5.5% underwent subsequent surgery for recurrent prolapse, with major intraoperative complications predominantly ureteric injury, plus single cases of small bowel injury and pelvic abscess requiring diverting colostomy.**

- At a mean of 5.1 years: **failure in one or more compartments 15.3%, but apical recurrence in very few.** De novo dyspareunia occurred. **Irritative voiding, obstructive voiding, SUI and overall urinary symptoms all improved.**
- Extraperitoneal bilateral suspension at 2 years: **apical recurrence 4.6%, global anatomical success 85.5%.** Mesh exposure occurred in 19.3%, arising from the concomitant anterior mesh reinforcement used in about half the cohort rather than from the suspension itself, and was managed conservatively or with minor intervention.
- At 10 years, with **objective recurrence defined as POP-Q stage 2 or worse and subjective recurrence as bulging symptoms: objective recurrence 19.1%, subjective 6.3%, reoperation 2.1%.** Combined anterior and posterior repair was protective; premenopausal status carried a fifteenfold increased risk of recurrence.
- **Suture material: permanent versus absorbable has not been shown to affect recurrence, and a randomised trial found absorbable non-inferior to permanent for apical recurrence.**

The OPTIMAL Trial

- **OPTIMAL, a randomised controlled trial of the Pelvic Floor Disorders Network, compared uterosacral suspension with sacrospinous ligament fixation.**
- **At 2 years, surgical success for uterosacral suspension was 64.5%, and results were similar for sacrospinous fixation.**
- **Perioperative pelvic floor muscle training was not associated with improved outcome.**
- **Counterpoint:** a single-institution study of stage 3 prolapse found **failure 25.7% after uterosacral suspension versus 7.8% after abdominal sacrocolpopexy.**

Concept — *The apex is the compartment this operation actually secures; the anterior compartment is its weakest result. Apical success approaches 98% while anterior success sits near 81%, so recurrence after a technically sound suspension is anterior by default — and the safe-suture rules exist because the ureter converges on the ligament from 4 cm to under 1 cm as it runs towards the cervix.*

Sacrospinous Ligament Fixation

- **First described by Richter in 1942, but not widely adopted until Nichols reported the technique in 1982.**
- Its appeal rests on **an extraperitoneal approach and a consistently strong structure on which to anchor the apex. Access was originally posterior; anterior access via the paravaginal space is also possible.**
- **Stated advantages:** success comparable to abdominal procedures; ability to repair concomitant defects; no laparotomy; shorter hospital stay; preservation of vaginal length and function; cost effectiveness. **These have not been fully re-evaluated in the era of laparoscopic and robotic repair.**
- **Meta-analysis:** both sacrospinous fixation and laparoscopic sacrocolpopexy restore apical support efficiently, but sacrocolpopexy may be favoured for durability and sexual function; in subanalysis there was no difference in recurrence between sacrospinous fixation and the laparoscopic route.
- **Structures at risk:** pudendal or inferior gluteal vessels; sciatic or pudendal nerves. Pudendal entrapment → posterior buttock pain radiating down the back of the thigh.
- **The signature disadvantage is alteration of the vaginal axis: unilateral fixation displaces the apex posteriorly and to the right, and this can produce anterior compartment recurrence even when an anterior repair has been performed.**
- **Unilateral right-sided placement is generally preferred, being technically easier for a right-handed surgeon. Bilateral anterior support gives a more midline apex, but there is little outcome evidence favouring bilateral over unilateral fixation.**

Surgical Anatomy of the Sacrospinous Ligament

- The sacrospinous ligament is approximately 7 to 8 cm long, running from the ischial spine laterally, medially beneath the coccygeus muscle, to insert into the sacrum.
- The pudendal nerve and vessels lie just proximal to their course around the ischial spine. Hypogastric venous plexus superiorly; haemorrhoidal vessels medially — retract carefully in both areas.
- To avoid the gluteal vessels, place the suture INTO the ligament, not behind it.
- Pudendal entrapment causes pain localising to the buttock or perineum; avoid it by placing sutures approximately 1.5 cm medial to the ischial spine.
- Branches from the S3 and S4 nerve roots are the likelier cause of postoperative gluteal pain.
- Three rules to avoid nerve entrapment: precise placement on the ligament; limited depth of needle penetration; no needle entry or exit above the upper extent of the ligament.
- The middle segment has the lowest density of associated nerves and arteries, and the nerves supplying the pelvic floor muscles are at higher risk of entrapment than the pudendal nerve.
- The optimal suture position is 1.5 to 3.0 cm medial to the ischial spine, placed directly into the ligament.
- Placing it too far medially risks failure of the repair, because the ligament fans and thins as it inserts into the sacrum, and there is a higher concentration of sacral nerves in that medial location.
- Cadaveric histology: nerve concentration is highest in the sacral portion and lowest near the ischial spine; the postulated link between embedded nerves and neuropathic pain could not be correlated clinically.
- Both the localisation and the depth of the suture influence sciatic neuralgia.

Reconciling the apparent contradiction

- Nerve density is lowest near the ischial spine, yet the clinical rule forbids suturing there. These are not in conflict: the danger at the spine is the pudendal neurovascular bundle passing around it, not nerves embedded within the ligament, while the danger medially is intraligamentous sacral nerve fibres plus a thin, fanned-out ligament. The middle segment avoids both.
- Uterosacral ligament: the intermediate portion. Sacrospinous ligament: the middle segment, 1.5 to 3.0 cm medial to the ischial spine. Both are safest in their middle third and dangerous at their bony ends.

Sacrospinous Ligament Fixation: Technique

- Mark the vaginal apex → reduce the apex to the intended ligament and tag that site for later fixation → choose the route by the concomitant compartmental defects → standard midline vaginal incision; separate the epithelium from the rectovaginal septum posteriorly, or the pubocervical fascia anteriorly →
- Anterior route: expose the paravesical space and sweep the tissue overlying spine and ligament medially. Posterior route: enter the perirectal space by bluntly mobilising the rectum medially → identify the ischial spine; with further dorsal and medial blunt dissection, palpate the ligament → continue until the rectum is retracted medially and the ligament adequately exposed →
- Heaney or Breisky–Navratil retractors, with lighted retractors for added visualisation → place the suture approximately one to two fingerbreadths medial to the spine, to avoid the structures in Alcock's canal (the pudendal canal) → place the sutures — two most commonly → perform any additional repairs at this point →
- Bring the sutures out of the vagina to the previously marked apex → if permanent suture is used, construct a pulley stitch → close the vaginal wall halfway before tying down, since the suspended apex

may be hard to reach → **tie so that the apex sits firmly against the coccygeal–sacrospinous complex, with NO suture material bridging a gap** →

- **Palpate the suture behind the vaginal cuff to confirm secure fixation** → leave the secured suture untrimmed until cystoscopy is performed → once bilateral efflux is confirmed, trim the anchoring suture and close the remainder of the incision.

Suture number, the pulley stitch and the devices

- **Bilateral fixation: one suture on each ligament. Unilateral fixation: two sutures placed 2 cm apart**, to prevent constriction of the vaginal apex.
- **Pulley stitch** — **purpose:** the knot remains internal and does not lie within the vaginal canal. **Method:** tie a knot onto the fibromuscular portion on the visceral side of the vagina to create a fixed point → pull the free end, suspending the apex to the ligament → secure with a definitive knot.
- **By palpation: Capio or Capio "Thin" suturing device. Under direct vision: Miya Hook, Autosuture Endo Stitch, Deschamps ligature carrier, or freehand needle driver passage.**

Figure 117.16 — Sacrospinous ligament fixation. With a unilateral suspension, the vagina is deflected to the right side and caudally. (From Richter K, Albrich W: Long-term results following fixation of the vagina on the sacrospinous ligament by the vaginal route. *Am J Obstet Gynecol* 141:811–816, 1981.)

Results of Sacrospinous Ligament Fixation

- The foundational series, including massive vaginal eversion and procidentia, reported **no recurrences of vault prolapse but a small number describing the vagina as too narrow postoperatively** — and the same signal appeared in the 1981 long-term series, where **a few patients were reluctant to resume sexual relations because of perceived narrowing.**
- **Global success 90.1%, apical success 95.7%** in a mixed series, the few failures being anterior compartment prolapse, with acceptable long-term results and minimal adverse effects.
- **Benson trial**, prospective randomised **bilateral sacrospinous fixation versus abdominal sacrocolpopexy: time to failure shorter with sacrospinous fixation (11 versus 22 months)**, and **cystocele reoperation three times more frequent in the vaginal group.** This is the trial underpinning the chapter's opening statement that the abdominal approach is more durable.
- A small series reported minimal blood loss, no transfusions, and **mild postoperative buttock pain resolving by 3 months.**
- **Excellent posterior and apical results but disappointing anterior results, with 16% having grade 2 or worse prolapse.**
- **Anatomical cure 96%** in one series, with intraoperative rectal and bladder laceration and haemorrhage requiring transfusion, and postoperative vaginal cuff abscess, fever and transient femoral neuropathy.
- **Cure of only 53% with 19% buttock pain** in the least favourable series quoted.
- At a median of over five years, **overall cure 84.1%**, with **preoperative stage 4 prolapse the only independent predictor of recurrence.**
- **Systematic review of 24 studies (1995–2011): overall cure 84.6%; apical recurrence 5.3%; anterior recurrence 18.3%.**
- Medium and long-term single-institution results: **recurrence 35% at 1 to 5 years and 20% beyond 5 years, and only 12% of those with recurrence reported any adverse effect on quality of life.**

The Anterior Compartment After Sacrospinous Fixation

- **The anterior compartment is a particularly vulnerable site for a new defect. Reported recurrent cystocele rates span 7.6–92%.**

- **The mechanism is vaginal retroversion — posterior displacement of the axis — and it may be evident intraoperatively as well as postoperatively.**
- **Many anterior defects remain asymptomatic and are never reoperated.**
- Pelvic Floor Disorders Network data on whether concomitant anterior repair helps after vaginal apical surgery: **in higher-stage prolapse (POP-Q point Ba greater than +1) undergoing sacrospinous fixation, concomitant anterior repair DID improve overall success; in lower-stage prolapse, and in all patients undergoing uterosacral suspension, it did NOT. Concomitant anterior repair earns its place only with high-stage anterior prolapse and only alongside sacrospinous fixation.**
- **Anterior sacrospinous fixation with anterior repair: anterior recurrence 9%, versus 24% after fixation with posterior repair. The bilateral anterior approach gave a low anterior recurrence rate of 7%, with theoretical advantages of improved capacity to withstand raised intra-abdominal pressure and less likelihood of rectal injury.**
- **Early healing predicts late success: absence of any compartment defect at 6 weeks conferred only a 3% likelihood of requiring further reconstructive surgery within 2 to 5 years.**

Complications of Sacrospinous Ligament Fixation

- **Ipsilateral gluteal pain in 15%, generally resolving spontaneously within 2 to 3 months when delayed absorbable sutures are used.**
- **Management of neuropathic pain:** observation, with counselling that symptoms may last up to 3 months; injection of the nerve with local anaesthetic where treatment is needed.
- **Rectal perforation is reported; recognised intraoperatively and repaired primarily, no sequelae followed.**
- **Injury to the pudendal nerve and internal pudendal vessels occurs with sutures placed too near the ischial spine.**

Concept — *The three-to-one ratio of anterior to apical recurrence is the defining outcome signature of this operation. Deflecting the apex posteriorly and to the right is what buys the strong extraperitoneal anchor, and the anterior compartment pays for it — which is why an anterior repair is added only where the anterior defect is already high-stage.*

Iliococcygeus Suspension

- **Advantage: maintains the vagina in a more normal axis. Principal indication: where the uterosacral ligaments are insufficient for attachment or cannot be identified. The anchoring site is the fascia of the iliococcygeus muscle.**
- **Approach as for uterosacral suspension and sacrospinous fixation, usually extraperitoneal → identify the iliococcygeus fascia lateral to the rectum and distal to the ischial spine → place the point of fixation 1 cm distal to the ischial spine, near the insertion of the ATRF →**
- **Pass the suture through the condensation of connective tissue at the vaginal apex, anteriorly through the pubocervical fascia and posteriorly through the rectovaginal fascia → place a matching suture contralaterally — bilateral fixation is recommended → tie securely; the vagina is suspended bilaterally → where permanent suture is used, apply the pulley stitch, tying the knot internally.**
- **Results: a minority experienced postoperative pelvic support loss, distributed across all three compartments, with the anterior compartment the commonest single site.**

Abdominal Sacrocolpopexy

- **Indications:** failed previous vaginal repair; isolated uterine prolapse and/or enterocele; younger women; highly active lifestyle; sexually active women; **those wanting one of the consistently most durable repairs and accepting a more invasive approach.**
- **It maximises functional vaginal length without significant distortion of the anatomical vaginal axis, and other pelvic floor defects and SUI can be addressed at the same operation.**
- **Robotic-assisted and pure laparoscopic approaches are now more common than open; the principles are identical regardless of approach.**

Open Abdominal Sacrocolpopexy: Technique

The two critical elements: permanent mesh (type I macroporous monofilament) as graft material, and secure fixation of that graft to the sacral promontory and the vaginal cuff.

Access and the promontory

- **Low lithotomy, giving both transvaginal and transabdominal access → Pfannenstiel or low midline incision → expose the true pelvis: release all pelvic adhesions, pack small intestine and sigmoid above the promontory, displace the sigmoid to the left →**
- **Incise the posterior peritoneum over the promontory, extending inferiorly along the RIGHT lateral aspect of the rectum towards the cul-de-sac → divide the fatty tissue with electrocautery → dissect carefully to avoid shearing the presacral veins, and avoid the middle sacral vessel, which traverses the promontory →**
- **Visualise the anterior surface of the promontory, identified by the anterior longitudinal ligament → place two to three interrupted non-absorbable monofilament sutures in the anterior longitudinal ligament, avoiding perforation of the midline sacral vessels — generally passed **under** the vessels and tied down **over** them; alternatively apply bipolar energy first where vessels overlie the ligament →**
- **Tag these sutures for later placement into the mesh. Tacking devices are an alternative, though minimal data confirm comparable efficacy.**

Culdoplasty

- **Place an end-to-end anastomosis (EEA) sizer or commercial vaginal stent in the vagina → identify the enterocele sac and secure it with an Allis clamp → if large, perform a Halban culdoplasty: linear permanent or delayed absorbable sutures through the posterior peritoneum and over the outer surface of the rectum to the vaginal cuff → **upward retraction of the sizer exposes the cul-de-sac; usually 4 to 6 sutures achieve closure.****
- **Halban uses LINEARLY placed sutures; Moschowitz uses a PURSE-STRING closure, which may predispose to ureteric angulation and obstruction.**
- **The culdoplasty may be omitted where an aggressive posterior dissection to the perineum is planned for an extended posterior mesh with retroperitonealisation.**

Vaginal dissection and graft attachment

- **Incise the peritoneum over the vaginal cuff and dissect it off the anterior vaginal wall → the extent of dissection varies widely: minimal dissection sufficient to fix mesh for 4 to 5 cm on each side, or extensive dissection lifting the posterior bladder wall and trigone off the vagina and continuing to the perineal body. Robotic practice has trended towards more extensive dissection with longer mesh segments, but no controlled study identifies the optimal technique →**
- **Mesh options:** prefabricated Y-shaped soft polypropylene, or sheets cut to preference; **a polypropylene, macroporous, monofilament, non-absorbable mesh is recommended → anterior segment**

approximately 2 cm wide by 6 cm long, sutured to the anterior vaginal wall with 6 to 8 permanent sutures (Prolene, Ethibond or Tyron) →

- Identify the bladder border to avoid suturing graft to bladder, and secure the lateral-most portions of the mesh to prevent folding → an EEA bowel sizer or vaginal obturator facilitates suture placement → either fold the graft over the cuff with the long end exiting posteriorly towards the sacrum, or fashion a T with the short arm on top of the vagina and the long arm secured to the lower vagina →
- Dissect the posterior peritoneum off the posterior vaginal wall, entering the rectovaginal space → this may be extended to the perineum to allow mesh attachment to the perineal body — the procedure then termed colpoperineopexy → attach the posterior mesh segment with 6 to 8 interrupted monofilament sutures, delayed absorbable or non-absorbable → the central culdoplasty sutures may be passed through the long arm of the mesh at this step.

Sacral fixation and closure

- Secure the graft to the promontory, avoiding excessive tension → set graft length by placing the obturator fully into the vagina WITHOUT pushing the vagina upward → lay the graft along the RIGHT lateral aspect of the rectum → leave two fingerbreadths between graft and rectum to prevent compression of the rectum over the graft →
- Bring the promontory sutures through the mesh and tie down securely → trim excess mesh only after sacral fixation → retroperitonealise: close the posterior peritoneum over the graft and cover the vaginal portion with the superior edge of the anterior peritoneum and bladder flap → perform any indicated procedure for SUI → cystoscopy to confirm ureteric patency → examine the patient to determine whether any ancillary transvaginal repair is needed.
- Tensioning — open versus minimally invasive: open uses the two-fingerbreadth tactile gap; robotic and laparoscopic cannot, and require a visual cue of the mesh attachment while extending the vagina cephalad with an EEA sizer to its maximal length without actively stretching it.

Figure 117.18 — Abdominal sacral colpopexy. (A) Incision in the peritoneal cuff overlying the vagina, exposing the vaginal cuff. (B) Two permanent sutures placed in the sacral promontory to secure the graft material. (C) Long incision in the posterior peritoneum, used to cover the graft material. (From Winters JC, Cespedes RD, Vanlangendonck R: Abdominal sacral colpopexy and abdominal enterocele repair in the management of vaginal vault prolapse. *Urology* 56(6 Suppl 1):55–63, 2000.)

Figure 117.19 — Abdominal sacrocolpopexy. Synthetic graft material sutured securely to the vaginal cuff with multiple interrupted permanent sutures. The peritoneal cul-de-sac is closed with linearly placed sutures to obliterate the potential space. Labelled: graft placement on vaginal cuff; enterocele incorporated in Halban sutures. (From Winters JC, Cespedes RD, Vanlangendonck R: Abdominal sacral colpopexy and abdominal enterocele repair in the treatment of vaginal vault prolapse. *Urology* 56(6 Suppl 1):55–63, 2000.)

Robotic-Assisted Laparoscopic Sacrocolpopexy: Technique

- Sacrocolpopexy may be performed laparoscopically or robotically. The basic steps are identical, differing only in method of abdominal entry, trocar placement, docking of the patient cart, and method of suture fixation. The da Vinci Surgical System is currently the only widely used robotic system in the United States.
- **Positioning:** patient on a Pink Pad or equivalent anti-slip device → general anaesthesia → low lithotomy in Allen stirrups; arms tucked; all bony prominences padded → pelvic examination; abdomen and vagina prepared → Foley catheter; nasogastric or orogastric tube.
- **Access:** entry open or closed to the surgeon's experience — closed entry offers quick access with low injury risk, but meta-analyses have not confirmed either superior → incision at the base of the

umbilicus; 8.5 mm robotic camera trocar → insufflate with carbon dioxide; diagnostic survey → two or three additional 8.5 mm robotic trocars at least 9 cm apart, semilunar or W configuration → an 8.5 to 11 mm laparoscopic trocar on the patient's RIGHT for the assistant → Trendelenburg; side- or parallel-docking preferred, preserving access to vagina, bladder and rectum.

- **Instruments:** robotic monopolar scissors, Maryland bipolar forceps, robotic bowel grasper.
- **Exposure:** adhesiolysis → sweep the bowel out of the pelvis and above the pelvic brim → identify the aortic bifurcation and iliac vessels; delineate the promontory at the midline → identify the ureters, particularly the RIGHT ureter, which lies close to the dissection → incise the peritoneum over the promontory, extending along the right lateral aspect of the rectum, creating peritoneal flaps that will later retroperitonealise the graft →
- **Divide the fatty tissue with electrocautery; avoid shearing the presacral veins; avoid the LEFT common iliac vein, which is not infrequently located more medial than the artery; avoid the middle sacral vessel → visualise the anterior longitudinal ligament → extend the peritoneal incision along the right pelvic sidewall into the pouch of Douglas.**

Vaginal dissection

- **Elevate and distend the upper vagina and apex with a vaginal obturator — large EEA bowel sizer, Lucite stent, or ColpAssist Vaginal Positioning Device → incise the peritoneum covering the vagina transversely; separate peritoneum and bladder from the anterior vaginal wall sharply and bluntly → extend this dissection approximately 6 to 8 cm caudally →**
- **Dissect just above the fibromuscular layer, in the loose areolar tissue of the vaginal wall — the correct plane is what prevents accidental entry into the vagina → if the wall is opened: irrigate copiously, then close in two layers with 2-0 or 3-0 delayed absorbable suture → redirect the apex anteriorly; open the peritoneum over the posterior wall → open the rectovaginal space bluntly to the rectal reflection, approximately 9 cm → for a sacrocolpoperineopexy, continue beyond the reflection to the perineal body.**
- **Why the plane matters: wrong plane → unintentional entry into the vaginal lumen → increased risk of future graft exposure, particularly with synthetic mesh.**

Graft fixation, tensioning and completion

- **Synthetic mesh should be lightweight, flexible and porous; most often a prefabricated Y-shaped mesh, tapered to preference.**
- **Switch to robotic needle drivers with a bowel-safe grasper; pass the mesh in via the assistant's trocar → secure the graft to anterior and posterior vaginal walls with delayed absorbable or permanent monofilament suture, or a combination → place sutures FULL THICKNESS without entering the vaginal lumen. Absorbable suture gives equivalent long-term outcomes to permanent for mesh fixation.**
- **Tensioning:** direct the obturator posteriorly towards the promontory and extend it maximally cephalad → release tension, allowing the apex to lie in a normal position with the prolapse reduced → avoid excessive tension — hypersuspension causes pain or de novo SUI. The suspension should reduce prolapse of the apex AND of the anterior and posterior vaginal wall segments.
- **Sacral fixation:** suture the graft tail to the anterior longitudinal ligament at the S1 vertebral body or at the level of the promontory. Too low — S3 or S4 — increases the risk of presacral venous plexus injury. Too high — the upper promontory — risks laceration of the middle sacral vessel or the LEFT common iliac vein. At S1 the middle sacral vessels are readily visible and can be isolated and avoided. Two to three permanent monofilament sutures.
- **Completion:** cystoscopy to exclude bladder injury, intravesical suture or mesh perforation and to confirm ureteric patency → close the peritoneum over the vaginal apex with delayed absorbable suture,

continuing to the promontory, completely retroperitonealising the graft → undock; out of Trendelenburg; desufflate; remove trocars under direct vision; close trocar sites.

Laparoscopic Sacrocolpopexy: Technique

- Anaesthetic and positioning considerations, abdominal access, camera trocar placement and adhesiolysis are similar to the robotic approach. A 0° laparoscope is typically used.
- **Port placement:** two additional trocars under direct vision in the right and left lower quadrants, lateral to the inferior epigastric vessels; one or two further 5 mm ports at the level of the umbilicus, lateral to the rectus muscle; additional 5 mm ports on the principal surgeon's side to permit two-handed operating.
- **Instruments:** monopolar scissors, bipolar cautery graspers, laparoscopic blunt-tipped dissector. Trendelenburg; bowel swept out of the pelvis. Further steps are nearly identical to the robotic procedure.
- Surgeon preference determines the needle holder and whether an open- or closed-ended knot pusher is used for extracorporeal tying. Some surgeons secure the mesh with barbed suture (V-Loc or Quill).
- The apical tail is sutured to the anterior longitudinal ligament at S1 with two or three permanent monofilament sutures.
- Titanium tacks are an alternative. Tack use may be associated with lumbosacral osteomyelitis and spondylodiscitis — rare but devastating complications — but most reported cases have in fact been related to SUTURES placed in the presacral space, not tacks.
- Cystoscopy, peritoneal closure, retroperitonealisation and closure are as for the robotic approach.

RESULTS OF SACROCOLPOPEXY

- Abdominal sacrocolpopexy maintains a functional vagina and restores maximal vaginal length and support; vaginal length was longer after sacrocolpopexy than after sacrospinous fixation.
- A non-absorbable monofilament synthetic mesh should be used. A randomised comparison of absorbable cadaveric fascia lata against monofilament polypropylene found **objective recurrence substantially more frequent with the fascial graft** — approximately one third versus under one tenth.
- Numerous authors report success rates above 90%.
- **Against vaginal apical repair:** abdominal repair gave **higher success with equivalent hospital stay**; at two years against sacrospinous fixation, **symptoms and anatomical success were equivalent but asymptomatic failures to the introitus and recurrent cystoceles were less frequent after the abdominal operation**. Vaginal procedures carry a higher rate of recurrent anterior defects even when a concomitant anterior colporrhaphy is performed.
- **The distal defect problem:** after an isolated abdominal repair that does not address concomitant pelvic floor defects, **recurrent cystocele and rectocele occur in up to one third**. Most remain asymptomatic, but secondary vaginal repair is required in up to 20%. Counsel preoperatively.
- **E-CARE:** at 7 years, POP recurrence increases progressively. Anatomical failure defined as POP-Q point C greater than -2, or Ba/Bp greater than +1. Across the two groups, **symptomatic failure 24–29%**; anatomical failure 22–27%; composite failure 34–48%. Half of the failures were asymptomatic, and half of these patients were never retreated.
- **Laparoscopic series:** one large series at a mean operative time of 97 minutes reported **96% satisfaction, no sexual dysfunction, POP recurrence 4%**, with isolated vaginal mesh exposure, bowel incarceration, spondylitis and mesh infection. A prospective series at a mean operative time of 180.5 minutes reported **objective cure 98% with an 18% failure rate concentrated in the posterior compartment** and no reoperation, plus **de novo SUI 7.4%, mesh erosion 4.5% all within one year, and mesh-related pain 2.7%**.

- **Pooled review of 11 robotic and laparoscopic series: objective cure 91%, satisfaction 92%, mean operative time 124 minutes (55–185), conversion 3% (0–11%).**

COMPLICATIONS OF SACROCOLPOPEXY

- **Intraoperative: haemorrhage or transfusion 4.4%; cystotomy 3.1%; enterotomy 1.6%; ureteric injury 1.0%.**
- **Postoperative: urinary tract infection 10.9%; wound infection 4.6%; ileus 3.6%; deep venous thrombosis or pulmonary embolism 3.3%; small bowel obstruction 1.1%.**
- **Presacral haemorrhage arises from disruption of the presacral vessels, is reduced by fixing the graft higher on the promontory, and may be prevented by pre-emptive bipolar cautery, particularly in minimally invasive surgery.**
- **CARE: reoperation for gastrointestinal complications 1.2%.**

Mesh exposure

- **Heralded by persistent pain, discharge or infection — vigilance in follow-up is required.**
- **Meta-analyses report vaginal mesh exposure in 3.4–5.4%, varying with mesh type; the extended CARE long-term follow-up gives 9.9% — the figure surgeons should quote.**
- **Symptoms develop on average 14 months after surgery, classically vaginal bleeding and discharge.**
- **Exposure is commoner with MICROPOROUS materials (Teflon, Gore-Tex) and less common with MACROPOROUS monofilament meshes.**
- **Management: transvaginal mesh excision with or without partial colpocleisis, or abdominal mesh excision.**

Hysterectomy at the time of sacrocolpopexy

- **Concomitant hysterectomy is suggested to increase mesh extrusion, but findings are not uniform and no randomised trial addresses the question.**
- **With meticulous two-layer closure of the vaginal cuff, exposure after concomitant hysterectomy was 0.8%, against 0% with a prior hysterectomy.**
- **Perform a meticulous two-layer cuff closure in a combined operation; strongly consider SUPRACERVICAL hysterectomy, which is preferred in conjunction with sacrocolpopexy; and inform the patient that vaginal hysterectomy with vault suspension is also effective for uterine prolapse.**

SACROCOLPOPEXY BY ROUTE: COMPARATIVE OUTCOMES

TABLE 117.9 CLINICAL OUTCOMES COMPARING ROBOTIC-ASSISTED (RASCP) VERSUS LAPAROSCOPIC (LSCP) AND/OR ABDOMINAL (ASCP) SACROCOLPOPEXY

Comparison and design	Follow-up	Objective success and definition	Operative time (min)	Other outcomes
Abdominal versus laparoscopic				
Randomised, post-hysterectomy vault prolapse	12 mo	Composite success 89.2% ASCP vs 83.8% LSCP; no apical recurrence of stage 2 or worse in either	—	Reoperation more frequent laparoscopically; sexual activity, dyspareunia and coital frequency improved in both; no significant difference overall

Comparison and design	Follow-up	Objective success and definition	Operative time (min)	Other outcomes
Multicentre randomised equivalence trial	1 yr	Point C met equivalence ($P > 0.050$); PGI-I much or very much better 90% ASCP vs 80% LSCP	131 vs 144 — no difference	Laparoscopic: lower blood loss, smaller haemoglobin fall, stay 3.2 vs 4.1 days ; more morphine used after ASCP but equal physical function at 1 wk
Robotic versus abdominal				
Retrospective, point C at 6 wk	6 wk	100 vs 100 (preoperative vs postoperative POP-Q); point C -9 vs -8	328 vs 255	Blood loss 103 vs 255 mL ; stay 1.3 vs 2.7 days ; fever 4.1% vs 0% ; conversion 1 vs 0%
Retrospective	44.2 mo	100 vs 100; 1 or fewer apical prolapse, point C -5 or better, total vaginal length ≥ 7	—	POP-Q and PFDI-20, PFIQ-7, PISQ-12 improved similarly by route (all $P > 0.05$)
Retrospective	15 vs 11.7 mo	94 vs 94; point C less than two thirds of total vaginal length, or Aa/Ba/AP/Bp 0 or less ; subjective 92 vs 96	—	Blood loss 90 vs 228 mL
Robotic versus laparoscopic				
Randomised, stage 2–4 post-hysterectomy apical prolapse	12 mo	88 vs 91 (preoperative vs postoperative POP-Q); anatomical and QoL outcomes identical at 6 and 12 months	227 vs 162 incision to closure (table gives 349 vs 280); docking only ~14 min; suture tying 98 vs 68	More pain at rest and with activity over 6 wk robotically, attributed to the right paracolic gutter accessory port, since changed to 8 mm; conversion 8 vs 5%
Retrospective, median stage 3	10.8 vs 3 mo	87.1 vs 91.3 at 3 mo	334 vs 325, not significantly different (table gives 234 vs 334)	Blood loss 50 vs 100 mL ; no difference in visceral injury or mesh erosion
Prospective	15 vs 18 mo	99 vs 99	215 vs 280	Conversion 15 vs 2%
Retrospective	3 mo	—	186 vs 176	Blood loss 48 vs 206 mL
Non-inferiority randomised	24 mo	100 vs 100	234 vs 192	Blood loss 56 vs 58 mL
Three-way, cohort and pooled				
Multicentre retrospective, RASCP vs LSCP vs ASCP	7.6 mo	94 vs 81 vs 91 (POP-Q stage 2 or better). Anatomical failure 14.3% minimally invasive vs 24.1% abdominal, predominantly posterior support ; robotic 5.9% vs abdominal 18.9%	206 minimally invasive vs 234 abdominal	Abdominal: higher blood loss (188 vs 122 mL), longer stay (2.8 vs 1.2 days), despite fewer concomitant procedures
Large retrospective cohort, uterosacral vs LSCP vs RASCP	—	Median time to failure 17.1 / 10.1 / 9.7 — months ; retreatment 14% / 8% / 23% ; retreatment almost entirely posterior (86% robotic, 67% laparoscopic) ; NO apical retreatments in either sacrocolpopexy group ; strict	—	Confounded: up to 40% of sacrocolpopexy patients had prior prolapse surgery . Adjusted, uterosacral had higher anatomical

Comparison and design	Follow-up	Objective success and definition	Operative time (min)	Other outcomes
		composite recurrence may reach 40–60% by year 6		recurrence but lower symptomatic bulge and retreatment
Systematic review (1157)	3–39 mo	No significant difference to 39 months; no subjective difference	—	Blood loss mean difference 95.5 mL less with robotic
Systematic review (2115)	3–60 mo	No significant differences in Ba, Bp, C	269 vs 229	Blood loss mean difference 86.5 mL less with robotic
Meta-analysis of robotic reoperation	—	Commonest reason for reoperation was recurrent prolapse, 3.3% (95% CI 1.4–5.2)	—	Majority posterior repairs for non-apical prolapse

- **Objective success is high and essentially route-independent** — 87% to 100% throughout, with both systematic reviews finding no significant difference in Ba, Bp or C.
- **The only consistent robotic advantage is blood loss**, quantified by both reviews at roughly 86 to 96 mL less.
- **Operative time consistently favours laparoscopy over robotics** where the two are compared directly, and the robot's time cost is suture tying, not docking.
- **The definitions of objective success are not comparable across studies** — the 100% figures reflect permissive definitions, not superior surgery. Conversion rates are reported inconsistently and are higher robotically where given.
- **Current position: widespread access to robotic technology has produced a concomitant decrease in strictly laparoscopic approaches; operative times are typically shorter laparoscopically, owing to docking time; anatomical outcomes are not significantly different.**

TABLE 117.10A AND B CLINICAL OUTCOMES AND COMPLICATIONS OF ROBOTIC-ASSISTED SACROCOLPOPEXY PATIENT SERIES

Design	Follow-up (mo)	Operative time (min)	Blood loss (mL)	Conversion (%)	Success (definition)	Complications
Prospective (31)	25	186	—	3	100 objective / 100 subjective (definitions not available)	Mesh erosion 0; bladder injury 7; other 3.2 — vaginal tear, port infection, ileus, myocardial infarction, syncope
Retrospective (52)	42	190	—	4	90 / 90 (definitions not available); 9.6% recurrent prolapse	Mesh erosion 0; bladder 2; bowel 2; other 1.9 — vaginal tear, UTI, wound infection
Prospective (120)	12	161	49	0	89 (POP-Q 0 or 1) / 94 (all points 0 or less, point C –5 or better); ultra-light Y mesh	Mesh erosion 0; de novo SUI 4; lower urinary tract symptoms 2; dyspareunia 2

- **Mesh erosion was zero in all three robotic series**, against the 3.4–5.4% meta-analytic and 9.9% extended CARE figures — but follow-up here is short.

- **Bladder injury is the commonest visceral complication of the robotic approach.**
- Pelvic Floor Distress Inventory (PFDI-20), Pelvic Floor Impact Questionnaire (PFIQ-7) and Pelvic Organ Prolapse–Urinary Incontinence Sexual Function Questionnaire (PISQ-12) all improved in the largest series.

Learning Curve

- **Laparoscopic sacrocolpopexy:** mean operative time fell from 196 to 162 minutes comparing the first 15 with the last 30 cases, with blood loss and complication rates unchanged. Approximately 30 to 40 cases are needed to master the operation, with an inflection point in operative duration after 18 to 24 cases.
- **Robotic sacrocolpopexy:** mean operative time fell from 222 minutes over the first 10 cases to 183 over the next 10, and the learning curve comprises about 50 cases.

Cost of Sacrocolpopexy

- **Cost may be calculated in several ways, so comparison across studies is difficult.**
- One cost-minimisation analysis found an overall 10% saving favouring robotic over abdominal (\$10,178 versus \$11,307), with the largest cost drivers the number of institutional robotic cases performed annually, length of stay, and cost per postoperative hospital day.
- Another, across all three routes, found robotic the MOST expensive (\$8508), then laparoscopic (\$7353), then abdominal (\$5792), with cost equivalence only when robotic operative time fell to 149 minutes.
- A randomised comparison found robotic cost \$1936 more than laparoscopic (95% CI \$417–\$3454, $P = 0.008$), driven mainly by operating room costs (\$1667, 95% CI \$448–\$2885, $P = 0.008$), excluding the initial robot purchase and yearly maintenance.
- **The two cost-minimisation analyses reach opposite conclusions; the reconciling variables are annual robotic case volume and operative time — the 149-minute threshold is the number to hold.**

Comparison of Apical Procedures

- **Cochrane review (2016), vaginal apical procedures versus sacrocolpopexy** — vaginal surgery carried a higher risk of awareness of prolapse (relative risk 2.11), repeat surgery (RR 2.28), recurrent prolapse (RR 1.89), SUI (RR 1.86) and dyspareunia (RR 2.53).
- **Important qualifier:** vaginal procedures nonetheless have high success rates with low reoperation rates, and may be the procedure of choice for many older patients.
- **No evidence of any difference in any primary outcome between the different types of apical vaginal native tissue repair.**
- Comparative effectiveness work found sacrospinous fixation and uterosacral suspension both effective, with similar anatomical and functional outcomes and few adverse events, and sacrospinous hysteropexy NOT INFERIOR to vaginal hysterectomy with uterosacral suspension for apical uterovaginal prolapse.
- More recent work goes further: sacrospinous hysteropexy is SUPERIOR to uterosacral suspension with vaginal hysterectomy, and superior to uterosacral hysteropexy.
- **But abdominal sacrocolpopexy ultimately achieves higher success than sacrospinous fixation (91.4% versus 88.3%) with lower dyspareunia (4.67% versus 14.36%).**

Concept — *Whichever route is used, the apex holds and the posterior compartment fails. Every comparative dataset locates failure and retreatment posteriorly, with essentially no apical retreatment after sacrocolpopexy — so route selection is a choice about morbidity, cost and operating time, not about whether the apex will be supported.*

COLPOCLEISIS

- **An obliterative procedure for post-hysterectomy vault prolapse or significant uterovaginal prolapse.**
Indication: properly counselled older patients who do not desire a functional vagina for sexual activity.
- **TOTAL colpocleisis:** for the post-hysterectomy patient. Epithelium removed to approximately 2 to 3 cm from the urethral meatus, with **COMPLETE** closure of the vaginal canal.
- **PARTIAL colpocleisis:** for uterovaginal prolapse with the uterus in situ — the Le Fort operation. Lateral channels are created to allow potential uterine drainage.
- **Advantages:** shorter operative time; regional anaesthesia or local anaesthesia with sedation feasible; minimal complications; minimal recurrence; decreased recuperative time.
- **Where the uterus is left in situ:** Papanicolaou smear, pelvic ultrasonography, and endometrial biopsy if indicated.
- **Hysterectomy at colpocleisis eliminates the risk of cervical and endometrial cancer, and eliminates the risk of PYOMETRA** — a serious complication of partial colpocleisis when the channels become obstructed. However, most patients selected for colpocleisis are older with significant comorbidity making concomitant hysterectomy undesirable — the reason the channels exist at all.

Partial Colpocleisis: Technique

- **Standard lithotomy, indwelling urethral catheter → perform any anti-incontinence procedure first → grasp the cervix with a tenaculum → mark a rectangular segment of epithelium on the anterior and posterior walls, from 3 cm from the urethral meatus to 3 cm from the cervix →**
- **Leave sufficient epithelium laterally to allow creation of the drainage channels → excise the epithelium → optionally place a 14 Fr red Robinson catheter or vessel loop along the sidewalls to assist in forming the channels →**
- **Starting at the leading edge of the prolapse, place successive layers of 2-0 delayed absorbable sutures in transverse rows** — linear sutures through the pubocervical fascia anteriorly and the rectovaginal fascia posteriorly → **as each row is tied the prolapse is reduced proximally; continue until it is above the levator plate and completely reduced →**
- **High perineorrhaphy:** excise a triangular segment of epithelium and approximate the fibromuscular tissues of the perineal body in the midline with absorbable suture, narrowing the introitus → **indigo carmine and cystoscopy to confirm ureteric patency.**

Total Colpocleisis: Technique

The differences from partial colpocleisis: no lateral channels, and the entire prolapsed vagina is denuded.

- **Denude the entire prolapsed vagina from the apex to 4 cm proximal to the urethral meatus → carry the dissection laterally to the posterior lateral sulcus → dissect the prolapse sac away from the epithelium** — the enterocele usually reduces easily and needs no further treatment →
- **Beginning at the leading edge, place circumferential PURSE-STRING sutures of 2-0 absorbable suture, incorporating pubocervical and prerectal (rectovaginal) fascia → tie each progressive suture, reducing the prolapse proximally, until adequately reduced → alternatively use transverse rows of interrupted sutures as for partial colpocleisis →**
- **High perineorrhaphy (levator myorrhaphy) to narrow the introitus and prevent recurrence → cystoscopy to confirm ureteric patency.**
- **The principle common to both: success depends on the amount of vaginal tissue sutured together. This creates a septum of support, enhanced by bringing the levator muscles together with the perineorrhaphy.**

- **It is unnecessary to open the enterocele sac if present, because the repair closes the potential space into which bowel might protrude. If the sac cannot be reduced, use purse-string absorbable sutures incorporating the sac and the uterosacral remnants.**

Results of Colpocleisis

- A 20-year series in which **the enterocele sac was deliberately left unopened** reported a **single vaginal eversion recurrence**, treated successfully by repeat colpocleisis, and **all but one patient expressly denied regret over an operation precluding intercourse.**
- At a mean of 2 years: **no recurrent prolapse, no complications, all satisfied**; mild SUI occurred only in those who had undergone **Kelly plication** for incontinence, and was treated successfully with collagen bulking.
- **Regret over loss of coital ability 12.9%** in the most quoted series.
- A multicentre prospective study at one year: **84% very satisfied or satisfied with the decision for vaginal closure**, only two dissatisfied, one repeat colpocleisis for recurrence, and **most did not experience worsening of body image.**
- Where three quarters had total colpocleisis: **success on the PGI-I in 91%**, corroborated on examination, with **urinary frequency and urgency the most bothersome residual symptoms**, in a third of patients.
- Systematic search 2000 to 2020, follow-up 30 days to 5 years: **over 90% of patients aged between 69 and 84; success 62.5–100% by POP-Q stage 1 or better, and 87.5–100% by no prolapse beyond the hymen; coital regret 0–12.9%; general decision regret 0–13.8%.** Colpocleisis is a heterogeneous procedure with high subjective and objective success, low coital regret and low complication risk.
- Review using validated questionnaires: **colpocleisis remains a viable option without compromising quality of life, body image or sexuality — provided patient selection is diligent.**
- Large population-based study: **overall complications during admission 9.6%, higher with concomitant hysterectomy (11.4% versus 9.5%, $P = 0.023$), and concomitant hysterectomy independently increased complication risk (odds ratio 1.93, 95% CI 1.45–2.57, $P < 0.001$).** Shared decision-making is essential, as these patients are older and more comorbid than those having non-obliterative surgery.

Results of Partial (Le Fort) Colpocleisis

- The original Le Fort series: **good anatomical results and symptom relief in the large majority; three recurrences, none reoperated**; immediate complications were one pulmonary embolism, fever and urinary tract infection.
- Another series: **no recurrences and no de novo SUI**, transient postoperative retention in a minority, and **no reported regret about sexual intercourse.**
- Over 300 patients: **anatomical success 98.1%, patient satisfaction 92.9%** — an effective, low-risk procedure for the appropriate elderly patient with advanced prolapse.
- Multi-institutional database: **43% of colpocleisis procedures performed on patients over 80; mean length of stay 2.3 days; mortality 1.5%; overall complications 6.8%; readmission 4.2%.**
- Multicentre and retrospective series: **improved body image, improved pelvic floor symptoms, low regret and high satisfaction after both partial and total colpocleisis.**

Urinary Considerations After Colpocleisis

- **Two mechanisms of postoperative incontinence: unmasking of occult SUI on reduction of the urethrovesical angle, and traction on the urethra when it is approximated to the posterior vaginal muscularis.**

- **The technical consequence:** spare at least 1.5 cm of the distal anterior vaginal wall adjacent to the urethral meatus.
- **All patients should be screened for occult SUI before colpocleisis, and an anti-incontinence procedure offered concomitantly to those with symptomatic SUI or occult SUI. In the asymptomatic patient the question remains controversial and unsubstantiated by prospective study.**
- **Frail elderly patients are at risk of urinary retention from impaired contractility — which may be identified preoperatively — and from other perioperative factors. This is harder to manage because these patients are less able to perform intermittent catheterisation. Discuss in advance; offer a suprapubic tube to those at significant risk.**

POSTERIOR COMPARTMENT REPAIR

Symptoms — three conceptual groups

- **Herniation symptoms:** vaginal bulging, or bleeding from excoriation of the epithelium.
- **Defecatory dysfunction:** stool trapping requiring vaginal splinting or manual digitation, defecatory urgency, constipation. **Defecatory dysfunction does NOT include faecal incontinence.**
- **Sexual dysfunction:** principally dyspareunia, with decreased desire and anorgasmia as contributors.

Distinguishing the Causes of Defecatory Dysfunction

- **Outlet obstruction** — from defects in the support of the posterior compartment, perineum and rectum, and from anismus. **Obstruction due to pelvic floor defects may be relieved by surgical restoration of anatomy.**
- **Motility disorders** — impaired transit of stool through rectum and anus; **treated with dietary modification and medication, not surgery.**
- **Anismus is failure of the puborectalis to relax during defecation, and often responds to biofeedback and pelvic floor physical therapy.**
- **In combined disorders, treat anismus or slow transit constipation — the commonest motility disorder — non-surgically BEFORE embarking on surgery.**

Who Benefits from Rectocele Repair

- **Predictors of benefit:** sensation of vaginal mass or bulge; need for digitation (splinting) to complete evacuation; non-emptying or partial emptying on defecography; a large rectocele.
- **Not predictive:** sensation of incomplete emptying and constipation — neither is specific to rectocele, and both may reflect irritable rectum or slow transit constipation.
- **What to tell the patient:** surgery will likely reduce vaginal protrusion symptoms and decrease or eliminate the need for splinting; **but constipation may persist**, because motility disorders and anismus coexist independently with prolapse and survive an otherwise successful repair.

Route — Transvaginal Versus Transanal

- A randomised comparison of **rectovaginal fascia plication versus transanal repair: symptom resolution was high with both (93% vaginal versus 73% transanal), but the vaginal approach gave better objective findings and a far lower recurrence rate (7% versus 66%).**

Historical Development

- **The traditional posterior colporrhaphy was devised in the 19th century to treat perineal tears sustained during vaginal delivery.**

- **The original description — plication of the pubococcygeus muscles and the posterior vaginal wall with reconstruction of the perineal body — was termed posterior colpoperineorrhaphy.** It produced a **rigid inferior shelf**, reduced posterior wall herniation, and helped prevent descensus of the vault or uterus.
- **Francis and Jeffcoate (1961): 50% incidence of dyspareunia after colporrhaphy with levator plication.**
- Later work reported **increased postoperative faecal incontinence, constipation, incomplete evacuation and dyspareunia** after traditional posterior colporrhaphy with levator plication.
- **Because of the postoperative dyspareunia, plication of the levator ani has largely been abandoned.** Site-specific repair and midline fascial plication **WITHOUT** levator ani plication are now the predominant surgical treatments for rectocele.

Posterior Colporrhaphy: Technique

- **Dorsal lithotomy.** If anterior or middle compartment repair is required, perform it **FIRST** → retract the anterior wall with a Heaney or Lone Star retractor with hooks → hydrodissection with saline or local anaesthetic, with or without adrenaline →
- **Midline incision through the posterior vaginal epithelium** — extending to the vaginal apex for a high rectocele, or starting at the most caudal position for smaller ones → **separate the rectovaginal fascia (muscularis) from the epithelium with Metzenbaum scissors, tips pointed TOWARDS the epithelium to avoid rectal injury** → dissect laterally until the pararectal attachments to the pelvic sidewall are visualised →
- **For large posterior defects, place a purse-string suture of 2-0 or 3-0 absorbable suture at the base of the rectal herniation** — this also draws the attenuated fascia together to aid reapproximation — **taking care not to foreshorten the posterior wall cephalad to caudad** →
- **Plicate the rectovaginal fascia in the midline with interrupted or continuous 2-0 absorbable sutures** — avoid excessive lateral placement or wide spacing, which cause overtightening or painful ridges → continue distally, incorporating the perineal body reapproximation → trim excess epithelium judiciously and close with running 2-0 absorbable suture → **perineorrhaphy before closing the epithelium if introital laxity dictates, avoiding overtightening.**

Site-Specific Repair: Technique

- **Devised to minimise the complications of posterior colporrhaphy.** Richardson described discrete defects in the rectovaginal fascia, identified both at operation and in cadaveric dissection.
- **The commonest defect is the TRANSVERSE configuration, separating the vaginal septum from the perineal body** — effectively a separation of the rectovaginal fascia from the perineal body. **The goal is to restore anatomy by closing these discrete defects, rather than plicating the midline.**
- **Midline incision on the posterior vaginal epithelium, as for posterior colporrhaphy** → separate the rectovaginal fascia from the epithelium through a virtually bloodless plane → insert the index finger of the non-dominant hand into the rectum to identify the fascial defects → close with interrupted absorbable sutures → trim excess epithelium judiciously and close with running absorbable suture.
- **The contrast: posterior colporrhaphy plicates the midline; site-specific repair closes discrete defects, identified by rectal examination during the operation.**

Perineorrhaphy: Technique

- **Excise a triangular incision at the mucocutaneous junction** → sharply remove the triangular island of posterior vaginal wall from the prerectal levator fascia → approximate the attenuated fibromuscular tissue of the perineal body with **HORIZONTAL MATTRESS** sutures → bringing these together reconstitutes the muscles of the urogenital diaphragm and restores support to the central tendon →

- **Create a SMOOTH CONTOUR along the suture line — ridges cause dyspareunia → vaginal packing moistened with saline, antibiotic solution, cream or gel may be placed if warranted.**

Results of Posterior Colporrhaphy

- Posterior repair with levator plication achieved **anatomical success**, but many patients reported **postoperative bowel symptoms and dyspareunia** — the adverse effects that led to abandonment of levator plication.
- A randomised trial of **posterior colporrhaphy versus site-specific repair versus site-specific repair with porcine small intestinal submucosa graft** found **similar functional and anatomical outcomes for colporrhaphy and site-specific repair**, no anatomical benefit from the porcine graft, and no difference in dyspareunia between the three groups.
- Follow-up of that cohort for bowel symptoms: **anatomical cure was associated with reduced postoperative straining and reduced sensation of incomplete evacuation**, but not with other bowel symptoms; overall, **incomplete emptying, straining, splinting and faecal incontinence improved significantly after rectocele repair**.
- Retrospective comparison in advanced posterior prolapse: **recurrence was HIGHER after site-specific repair than after midline plication — 33% versus 14% for second degree, and 11% versus 4% for third degree**; **symptomatic rectocele recurrence likewise greater after site-specific repair (11% versus 4%)**, with de novo dyspareunia and bowel symptoms the same in both groups.
- Prospective study of rectovaginal fascial plication: **improved anatomical outcome correlated with improved function**; **87% no longer had obstructive defecation**; significant improvements in awareness of prolapse, obstructive defecation, straining, hard stools, dyspareunia and digitation; **dyspareunia fell from 37% to 5%**.
- Another prospective fascial plication series: **87% improvement in bowel symptoms at 6 weeks**; **vaginal symptom relief 73–92% at 18 months**; **65% improvement in defecatory symptoms**; **38% improvement in sexual discomfort**; **no de novo dyspareunia or bowel symptoms**.

Site-specific repair series

- Prospective series: the great majority had a genuinely discrete defect, a small minority only tissue attenuation. **85% of those with preoperative bowel symptoms reported resolution at 3 months**; **de novo dyspareunia 3%**.
- Retrospective series: improvement or cure for pain or pressure, vaginal mass, splinting and difficulty with defecation — but **CONSTIPATION DID NOT CHANGE SIGNIFICANTLY**. **Preoperative dyspareunia improved in 73%, worsened in 19%**, and arose de novo in a small number. Emotional health improved, specifically thoughts of embarrassment and frustration.
- Site-specific repair with reattachment of the rectovaginal septum to the perineal body in all: **significant relief of protrusion, manual evacuation, difficult defecation and dyspareunia**; **constipation unchanged**. **At one year, roughly two thirds of those who used manual evacuation preoperatively had returned to doing so — attributed to functional decompensation of the rectum, which surgery does not correct**.
- Discrete fascial defect repair: **86% had prolapse stage 1 or better at 12 months**; **splinting eliminated in 63% of those reporting it**; **mean satisfaction 8.6 out of 10**. Mean improvement did NOT correlate with anatomical correction but DID correlate with relief of defecatory symptoms.

Guideline position

- **Fifth International Collaboration on Incontinence: a GRADE B recommendation, on level 1 and 2 evidence, that midline plication posterior repair WITHOUT levatorplasty may have superior objective**

outcomes to site-specific repair. The committee concluded that midline fascial plication without levatorplasty is the procedure of choice for posterior compartment prolapse.

- **Counterpoint: the Cochrane review (2018) concluded that, given the paucity of data and lack of good trials, the evidence was INSUFFICIENT to determine the relative effectiveness of the two procedures.**

Biological Interposition Graft Repairs of the Posterior Compartment

- **Grafts have been used, but outcomes are not significantly better than traditional repair. Several authors caution against SYNTHETIC materials in the posterior compartment because of dyspareunia and visceral erosion, favouring biological grafts.**
- **Site-specific repair with cadaveric dermis, graft fixed proximally to the vaginal apex, laterally to the levator ani, distally to the perineal body: 93% success by POP-Q at an average of 13 months.**
- **Cadaveric fascia lata: 83–89% symptomatic improvement; 90% of those examined had a grade 0 rectocele by the Baden–Walker classification.**
- **PelviSoft, a fenestrated porcine dermal acellular matrix: no wound complications.**
- **Against these encouraging results: at 3 years, 41% had recurrence of stage 2 or worse after collagen mesh for symptomatic rectocele, and more than half reported incomplete rectal evacuation; the authors advocated further study before routine use.**

Cochrane review (2018) — traditional versus biological graft repair

- **Awareness of prolapse: insufficient evidence of any difference (relative risk 1.09, 95% CI 0.45–2.62). Repeat surgery for any prolapse: insufficient evidence (RR 0.60, 0.18–1.97). Recurrent posterior wall prolapse: no conclusive difference (RR 0.55, 0.30–1.01).**
- **Postoperative complications were MORE common with biological repair (RR 1.82, 1.22–2.72) — the only statistically significant finding of the four, and it is against grafts.**
- **Counselling figures: graft extrusion 1.3–12%; dyspareunia may reach 60%; biological graft options are at present quite limited.**

KEY POINTS: POSTERIOR COMPARTMENT REPAIR

- **Because motility disorders often coexist with posterior compartment prolapse, counsel patients that constipation may persist despite successful anatomical correction.**
- **Levator ani plication should NOT be performed during rectocele repair, because of the high risk of postoperative dyspareunia.**
- **Repair of the perineal body is often performed at the time of rectocele repair, to recreate approximation of the rectovaginal connective tissues to the central tendon of the perineum.**
- **Use of reconstructive materials, synthetic or biological, to augment posterior vaginal wall repair is NOT supported by the body of evidence, although some smaller trials report conflicting outcomes.**

Concept — Constipation is the symptom that does not improve. Across every series, anatomical repair relieves bulge, splinting and straining while leaving constipation unchanged, because motility disorders and anismus are independent of prolapse — which is why the operation is selected on bulge and splinting, and why satisfaction tracks defecatory symptom relief rather than POP-Q.